
Software Requirements Specification

For Student Performance Visualization Tool

Version 1.0

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Revisions

Version	Primary Author(s)	Description of Version	Date Completed
0.1	Group 39	Statement of Work (SOW) prepared and project scope finalized.	Feb 9
0.2	Group 39	Week 1 Progress – Requirements analysis, system design, and data structure preparation	Feb 16
0.3	Group 39	Week 2 Progress – Login and user authentication pages implemented.	Mar 2
0.4	Group 39	Week 3 Progress – Implementation of data input and performance calculation modules.	Mar 9
0.5	Group 39	Week 4 Progress – Completed Software Requirements Specification (SRS) documentation.	Mar 16
0.6	Group 39	Implementation of student search, class average calculation, and performance comparison features.	Apr 6
0.7	Group 39	User interface integration completed and full system functionality achieved.	Apr 30
0.8	Group 39	Basic data security features implemented and system prepared for deployment.	May 5
1.0	Group 39	Final project submission including working application, dataset, documentation, and final project report.	May 12

1 Introduction

The **Student Performance Visualization Tool** is a software system designed to help teachers and students analyze academic performance using graphical visualizations. The application collects student academic data such as subject marks, attendance, and grades, and presents this information through charts and dashboards. By visualizing performance metrics like total marks, average scores, and percentages, the system helps users easily understand academic progress and identify strengths and weaknesses in different subjects.

This section of the document provides an overview of the purpose, scope, audience, and conventions used in the Software Requirements Specification (SRS). It introduces the main objectives of the project and explains how the system supports better understanding of student performance through clear data visualization and analysis.

1.1 Document Purpose

This Software Requirements Specification (SRS) document describes the requirements for the **Student Performance Visualization Tool**. The purpose of this document is to define the functional and non-functional requirements of the system, including system features, user interactions, and system constraints.

This document provides a clear description of how the system will collect student academic data, process information, and generate visual reports such as charts and dashboards to help teachers and students analyze academic performance.

1.2 Product Scope

The **Student Performance Visualization Tool** is a web-based system designed to collect, analyze, and visualize student academic data. The system allows teachers to upload or enter student marks, attendance, and academic information.

The system processes this data and generates visual insights using charts and dashboards such as bar charts, pie charts, and line graphs. These visualizations help both teachers and students easily understand academic performance and identify strengths and weaknesses.

The main objectives of the system include:

- Managing student academic records
- Calculating performance metrics such as total marks, average, and percentage
- Generating visual reports using charts and graphs
- Helping teachers monitor class performance

- Helping students track their academic progress

1.3 Intended Audience and Document Overview

This document is intended for the following readers:

- **Developers** – Developers will use this document to understand system requirements and implement the application according to the specified features.
- **Project instructors and evaluators** – to review the project design and functionality.
- **Users (Teachers and Students)** – Understand the system features and how the system supports academic performance analysis.

The document is organized as follows:

- Section 1 introduces the system and its purpose.
- Section 2 provides an overview of the system and its functions.
- Section 3 describes the detailed functional requirements of the system.
- Section 4 explains non-functional requirements such as performance and security.

1.4 Definitions, Acronyms and Abbreviations

API	Application Programming Interface used for communication between frontend and backend
SRS	Software requirements Specification
UI	User Interface
UX	User Experience
CSV	Comma Separated Values file used for storing student data
DB	DataBase
Node.js	Backend runtime environment
Chart.js	Library used for visualizing student performance

1.5 Document Conventions

This SRS follows the IEEE Software Requirements Specification (IEEE 830/29148) format. The document is written using consistent formatting to ensure readability and clarity.

Formatting conventions used include:

- Font: Arial
- Font size: 11 or 12
- Line spacing: Single
- Margins: 1 inch
- Section titles follow numbered headings (1, 1.1, 1.2 etc.)
- Italics are used for comments or notes

These conventions ensure that the document remains clear and consistent.

1.6 References and Acknowledgments

The following resources were used while preparing this SRS:

- *IEEE 830 Software Requirements Specification Standard*
- *Google Maps Platform Documentation*
- *Mapbox API Documentation*
- *Node.js Official Documentation*
- *PostgreSQL Documentation*
- *Software Engineering Course Guidelines – Mahindra University*

The authors acknowledge the course instructor and teaching assistants for their guidance during the development of this project.

2 Overall Description

2.1 Product Overview

The **Student Performance Visualization Tool** is a web-based application that helps teachers and students analyze academic performance through graphical representations. The system follows a **three-tier architecture consisting of presentation layer, application layer, and data layer**.

Teachers can enter student marks, attendance, and academic information into the system. The system processes this information and generates visual dashboards that show student performance trends, class averages, and subject-wise comparisons. Students can log into the system to view their individual performance reports, including subject marks, attendance, and grade summaries.

2.2 Product Functionality

The system provides the following major functionalities:

- User login and authentication for teachers and students
- Entry and management of student marks and attendance
- Calculation of total marks, average marks, and percentage
- Generation of student grades based on performance
- Visualization of student performance using charts and graphs
- Display of class average and performance comparison
- Identification of top-performing students
- Identification of weak areas in subjects
- Display of attendance records and statistics

2.3 Design and Implementation Constraints

The system development is subject to the following constraints:

- The system must follow **three-tier architecture**.
- UML diagrams must be used to model the system architecture and use cases.
- The frontend must be developed using **HTML, Tailwind CSS, and JavaScript**.
- The backend must be implemented using **Node.js and Express**.
- The system will use **MongoDB or MySQL** for storing academic data.
- Data visualization must be implemented using **Chart.js or D3.js**.
- The project must be completed within the academic semester timeline.

2.4 Assumptions and Dependencies

The following assumptions are made for the system:

- Student academic data will be available in structured formats such as **CSV or JSON files**.
- Users will have internet access to use the web application.
- Teachers will provide accurate student marks and attendance information.
- The system depends on chart visualization libraries to generate performance reports.

3 Specific Requirements

3.1 External Interface Requirements

3.1.1 User Interfaces

The system provides a web-based user interface that allows teachers and students to interact with the system.

- *Login page*
- *Marks entry form*
- *Student data management page*
- *Dashboard showing academic performance*
- *Charts and performance reports*
- *Subject-wise analysis*

The user interface will include charts and graphs to visualize academic performance.

3.1.2 Hardware Interfaces

The system interacts with the following hardware components:

- Personal computers and laptops used to access the web application
- Smartphones or tablets for mobile browser access
- Internet connectivity hardware required to access the server

3.1.3 Software Interfaces

The system interacts with the following software components:

- Node.js backend server for processing application logic
- MongoDB or MySQL database for storing student records
- Chart.js library for visualizing student performance
- Web browsers such as Chrome, Edge, or Firefox for accessing the application

3.2 Functional Requirements

The system must provide the following functional capabilities.

F1: The system shall allow teachers and students to log into the system using valid credentials.

F2: The system shall allow teachers to enter student academic marks and attendance records.

F3: The system shall store student academic data in the database.

F4: The system shall calculate total marks for each student.

F5: The system shall calculate average marks and percentage.

F6: The system shall assign grades to students based on their performance.

F7: The system shall generate graphical visualizations such as bar charts, pie charts, and line graphs.

F8: The system shall display class performance statistics such as class average and grade distribution.

F9: The system shall allow students to view their individual performance reports.

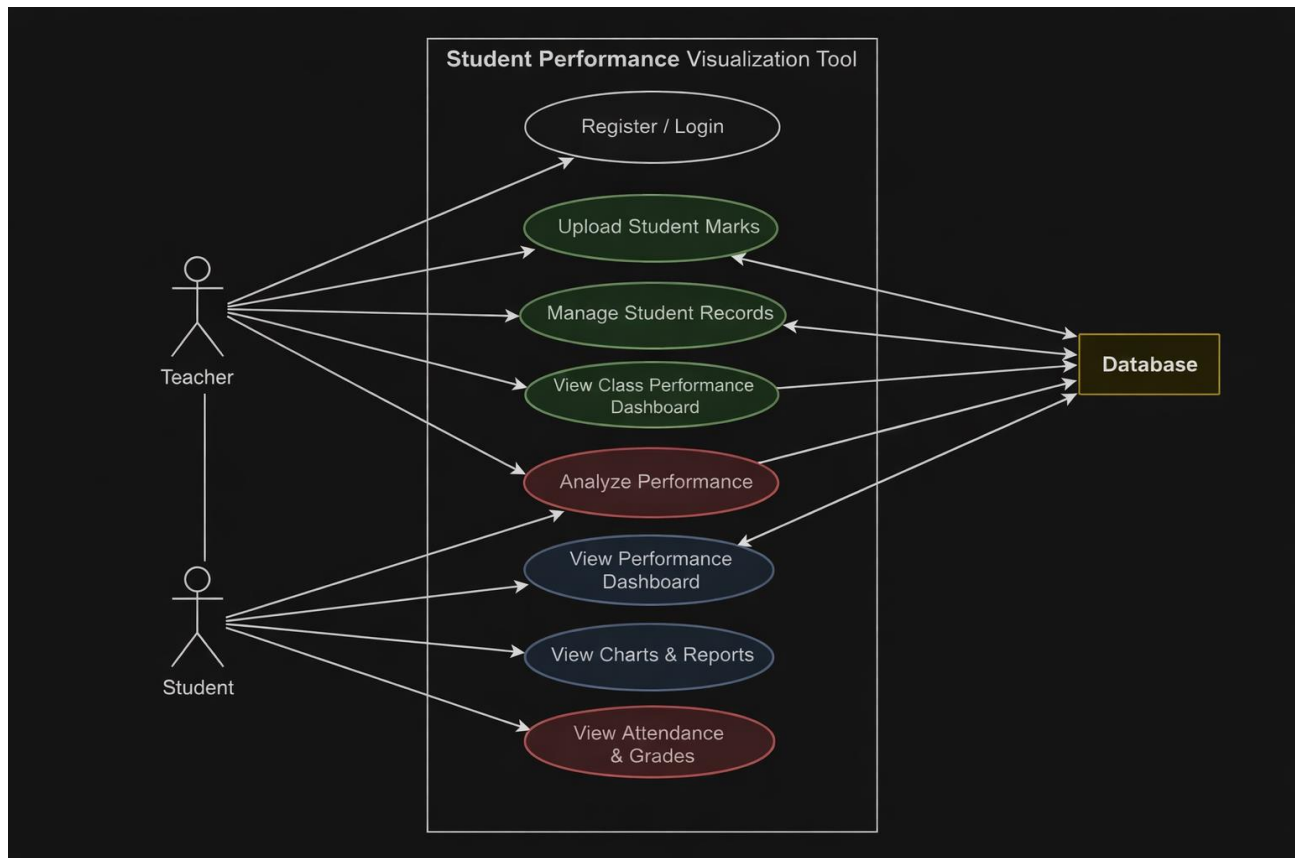
F10: The system shall identify subject strengths and weaknesses for each student.

F11: The system shall display attendance information for students.

F12: The system shall allow teachers to view all student performance records.

3.3 Use Case Model

The use case model describes the interaction between the users and the Student Performance Visualization Tool. The primary actors in the system are **Teacher and Student**. Teachers interact with the system to enter student marks, while students interact with the system to view performance reports and visualizations.



3.3.1 Use Case U1 - Plan Commute

Author: Group 39

Purpose: The purpose of this use case is to allow teachers to enter and store student marks in the system.

Requirements Traceability: F3, F4, F5, F6, F7, F8

Priority: High

Preconditions:

- The teacher must be logged into the system.
- The system must be connected to the database.

Postconditions:

- Student academic data is stored in the database.
- Performance metrics such as total marks, average, and percentage are calculated.

Actors:

- Teacher

Extends: None

Basic Flow

1. Teacher logs into the system.
2. Teacher opens the marks entry module.
3. Teacher enters student marks for different subjects.
4. Teacher submits the data.
5. The system stores the student academic data in the database.
6. The system calculates performance metrics.

Alternative Flow

If any required field is missing, the system displays an error message and asks the teacher to enter valid data.

Exceptions

If the database connection fails, the system displays an error message indicating that the data cannot be saved.

Includes: U2: View Performance Dashboard

Notes/Issues: The system should ensure that duplicate student records are not created.

3.3.2 Use Case U2 – View Student Performance Dashboard

Author: Group 39

Purpose: This use case allows students to view their academic performance and graphical analysis through the dashboard.

Requirements Traceability: F6, F7, F9

Priority: High

Preconditions:

- Student must be logged into the system.
- Student performance data must exist in the database.

Postconditions:

- The system displays charts and reports representing student academic performance.

Actors:

- Student

Basic Flow

1. Student logs into the system.
2. Student opens the performance dashboard.
3. The system retrieves the student performance data.
4. The system generates visual charts such as bar charts, pie charts, and line graphs.
5. The system displays the performance dashboard.

Alternative Flow

If no performance data is available, the system displays a message indicating that no records exist.

Exceptions

If the system fails to retrieve data from the database, an error message is displayed.

4 Other Non-functional Requirements

4.1 Performance Requirements

*P1: The system shall process student data and generate performance visualizations within **2 seconds** after the user requests the dashboard.*

P2: The system shall support multiple users accessing the system simultaneously.

P3: The system shall load the dashboard within 2 seconds after login.

P4: The system shall allow teachers to enter and save student marks quickly without delays.

These requirements ensure that the system works smoothly and responds quickly when users interact with important features.

4.2 Safety and Security Requirements

The system must ensure the **security and protection of student academic data**.

S1: The system shall require user authentication before accessing dashboards.

S2: User credentials shall be securely stored in the database.

S3: Only authorized teachers shall be allowed to modify student marks and attendance.

S4: Students shall only be able to view their own performance data.

S5: The system shall display appropriate error messages when invalid data is entered.

4.3 Software Quality Attributes

4.3.1 Reliability

The system should operate reliably without frequent failures.

- *The system shall accurately store and retrieve student academic data.*
- *The system shall prevent loss of data during system operations.*
- *The system shall maintain consistent system functionality during normal operation.*

4.3.2 Maintainability

The system should be designed in a way that allows future improvements and updates.

- *The system architecture should follow modular design principles.*
- *Developers should be able to modify system modules without affecting the entire system.*
- *The code should be well documented for future maintenance*

4.3.3 Usability

The system should be easy to use for teachers and students.

- *The user interface should be simple and intuitive.*

- *Teachers should be able to easily enter student marks through clear forms.*
- *Students should easily understand their performance through graphical visualizations such as charts and dashboards*

4.3.4 Portability

The system shall support modern web browsers including:

- *Google Chrome*
- *Microsoft Edge*
- *Mozilla Firefox*

The system should also be accessible from different devices such as laptops, desktops, and tablets.

4.3.5 Availability

- *The system should be available to users whenever required.*
- *The system should be accessible through web browsers at any time.*
- *The system should minimize downtime during system maintenance.*

5 Other Requirements

The system must maintain accurate academic performance records for students. Teachers must ensure that student marks and attendance information entered the system are correct and up to date.

The system should support importing student data in structured formats such as **CSV or Excel files**, allowing teachers to upload student academic records easily.

Appendix A – Data Dictionary

Entity	Attributes
Users	user_id, username, password, role
Students	student_id, name, attendance
Marks	subject1, subject2, subject3, subject4, subject5
Performance	total_marks, average_marks, percentage, grade

Appendix B - Group Log

Date	Activity
Feb 9	Team meeting to discuss project objectives and prepare the Statement of Work (SOW) .
Feb 16	Week 1 Progress – Requirements analysis, system design, and data structure preparation
Mar 2	Week 2 Progress – Login Page Implementation. Implemented the user authentication interface including login and registration pages.
Mar 9	Week 3 Progress – Implementation of data input and performance calculation modules
Mar 16	Week 4 Progress – Software Requirements Specification (SRS). Completed the SRS document including system requirements, use cases, and architecture description.